DOORDASH × NOVA HACKATHON

Smart Kitchen

Sophie Zhao

FastAPI & SQLite Engine | OpenRouter Structured LLM | QR Handoff Validation

THE CHALLENGE

Two Worlds That Barely Speak to Each Other

Front House: Peak Execution

Busy rushes cause staff to miss sold-out dishes

Orders keep arriving for items that can't be made.

- **Out-of-Stock Orders:** Forced cancellations break patron trust.
- **Handoff Bottlenecks:** Drivers scramble through identical paper bags.

Back House: Operational Blindness

Margins sit at 3–5%.

Inventory is difficult to track in receipts, memories, and delivery notes.

- **Manual Counts:** No real-time deduction by ingredient.
- **Late Awareness:** Cooks realize patties are gone mid-rush.

SYSTEM ARCHITECTURE

The Three Synchronized Databases

DATA STORE 01

Supplier Intake

Logs every shipment delivered, package sizes, and unit prices.

DATA STORE 02

Live Inventory

Real-time balances tracked ingredient-by-ingredient with reserve caps.

DATA STORE 03

Recipe Catalog

Maps each dish to specific ingredients and quantities of each ingredient

Curl

```
curl -X 'POST' \
  'http://127.0.0.1:8000/shipments' \
  -H 'accept: */*' \
  -H 'Content-Type: application/json' \
  -d '{
  "ingredient": "beef patty",
  "quantity": 1,
  "unit_price_cents": 150
}'
```

Request URL

```
http://127.0.0.1:8000/shipments
```

Response body


```
[
  {
    "ingredient": "beef_patty",
    "unit": "patty",
    "quantity_on_hand": 6,
    "reserve_threshold": 0,
    "total_received": 6,
    "last_unit_price_cents": 150
  },
  {
    "ingredient": "burger_bun",
    "unit": "bun",
    "quantity_on_hand": 10,
    "reserve_threshold": 0,
    "total_received": 10,
    "last_unit_price_cents": 40
  },
  {
    "ingredient": "cheese_slice",
    "unit": "slice",
    "quantity_on_hand": 12,
    "reserve_threshold": 0,
    "total_received": 12,
    "last_unit_price_cents": 25
  },
  {
```


esponse body


```
[
  {
    "id": 1,
    "name": "Classic Burger",
    "price_cents": 950,
    "is_available": true
  },
  {
    "id": 2,
    "name": "Cheeseburger",
    "price_cents": 1050,
    "is_available": true
  },
  {
    "id": 3,
    "name": "Double Cheeseburger",
    "price_cents": 1300,
    "is_available": true
  },
  {
    "id": 4,
    "name": "Chicken Sandwich",
    "price_cents": 1000,
    "is_available": true
  },
```

WORKFLOW: CUSTOMER STOREFRONT

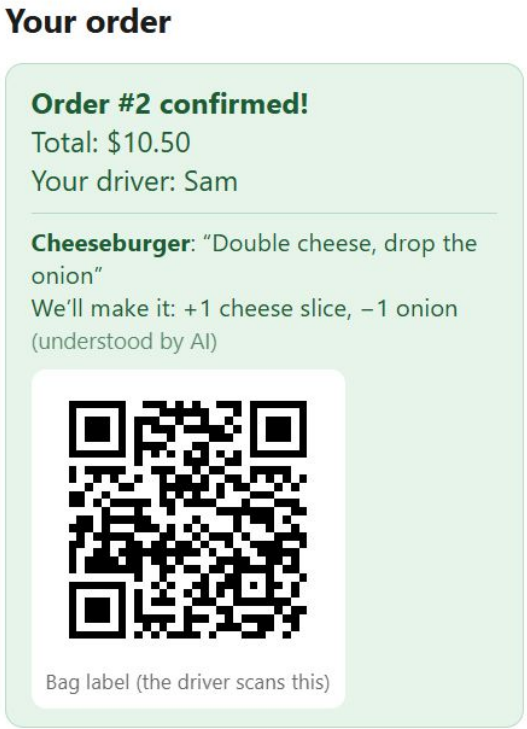
Translating Unclear Notes into Operational Data

> *"Example: A customer orders a Cheeseburger with note: 'double cheese, drop the onion'."*

1. OpenRouter LLM Call: Evaluates custom instructions into strict JSON deltas (+1 cheese, -1 onion).

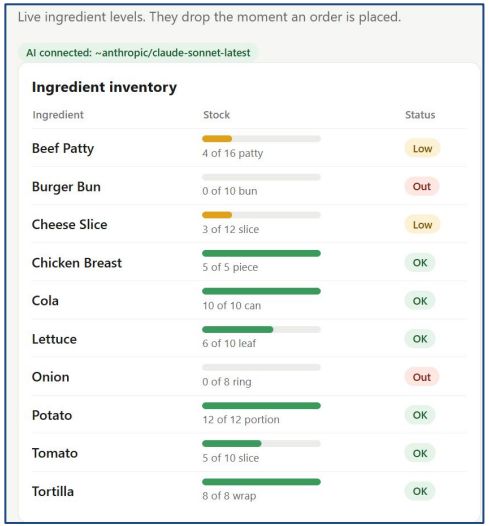
2. Verified Deduction: Checks real stock before mutating inventory in Database B.

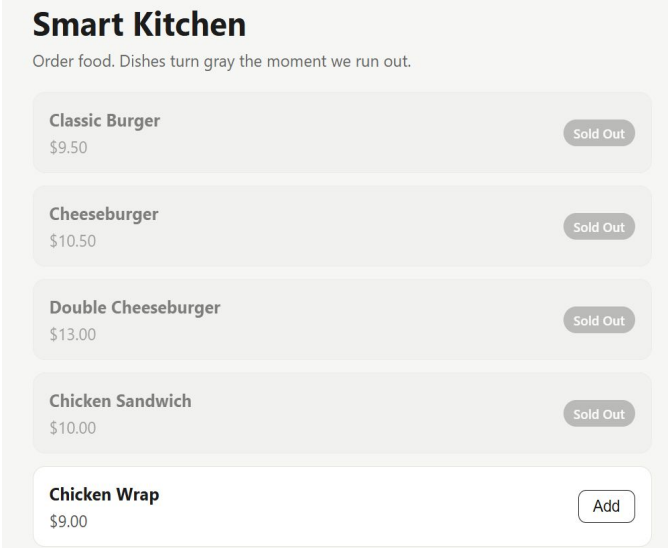
3. Digital Receipt: Generates QR code on the printed bag label.



WORKFLOW: KITCHEN DASHBOARD

Not Enough Ingredients? Dish Grayed Out



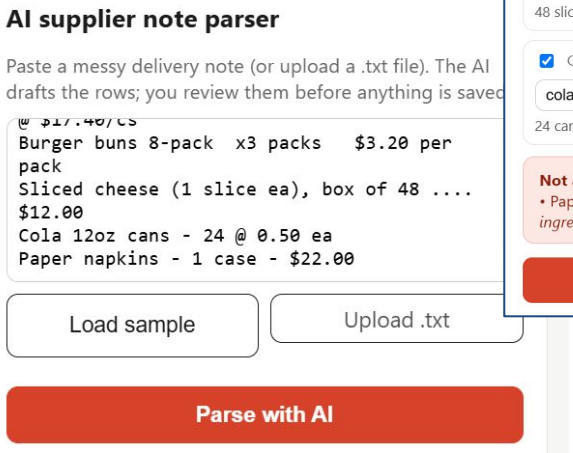


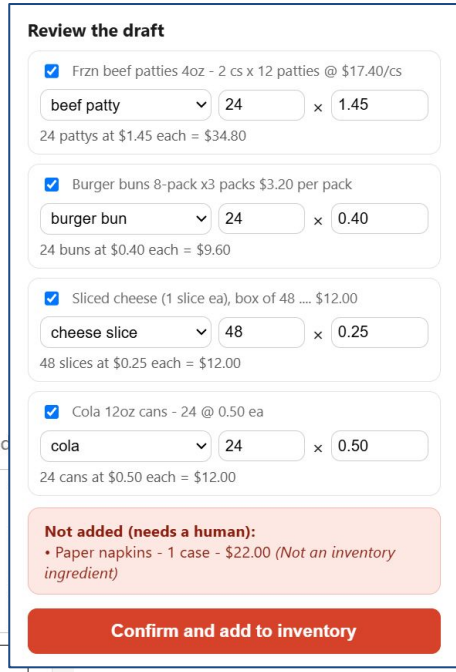
BACK OF HOUSE: INTAKE

AI Manifest Parser with Human Control

From Scribbles to Clean Inventory

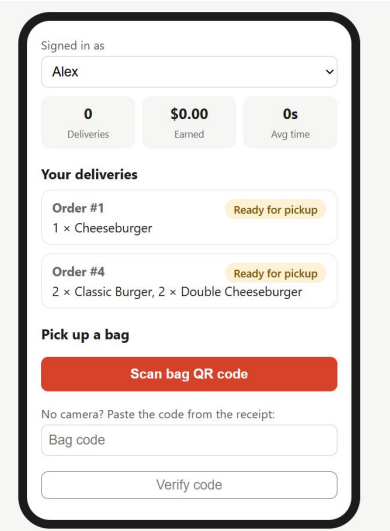
- **Cost-per-Unit Calculation:** Computes exact whole-cent costs to maintain precision.
- **Supply Discrimination:** Flags non-food items (e.g., paper napkins) in red without contaminating recipe databases.
- **Human Approval Gate:** The AI only drafts; stock updates only when an employee clicks *Confirm*.





FRONT OF HOUSE: DRIVER VERIFICATION

Two-Step Verification Protocol

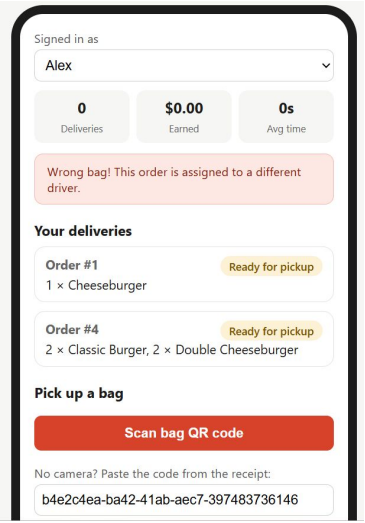


Step 1: Pickup Bag Verification

Driver scans bag QR code. If mismatched, the server rejects it with a **403 Forbidden** before the driver walks out.

Step 2: Customer Drop-off

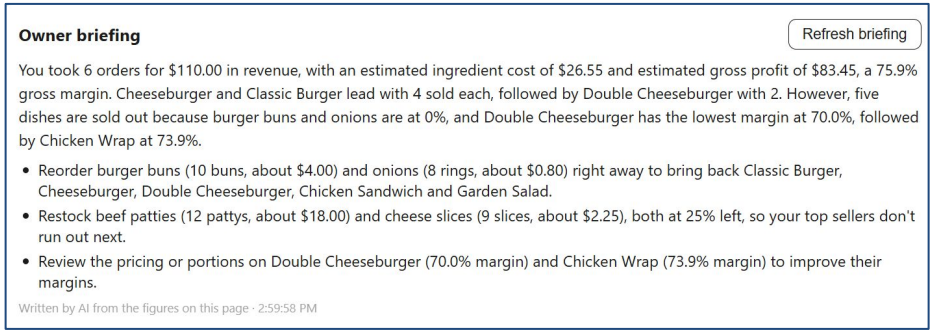
Drop-off confirmation is locked until physical verification occurs. Performance metrics and payouts reflect verified handoffs.

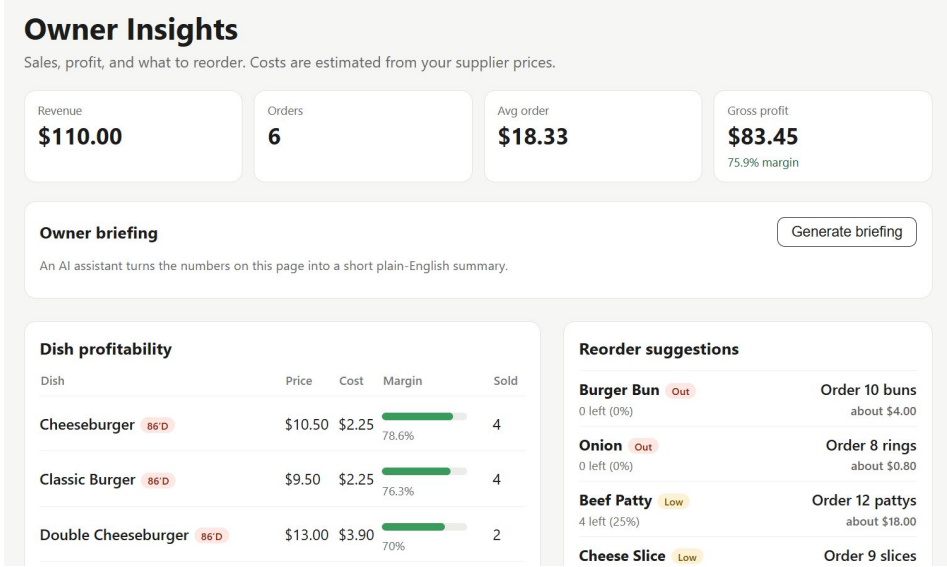


BACK OF HOUSE: FINANCIAL VISIBILITY

Owner Briefings

- **Item Profitability:** Live gross margin bars across all dishes.
- **Reorder Forecasting:** Proactive alerts before ingredients run empty.
- **Natural Language Briefing:** One-click AI synthesis gives clear operational advice without an accountant.





TECHNICAL DIFFERENTIATORS

Disciplined, Production-Grade AI Engineering

Translator, Not Boss

Structured JSON outputs strictly validated via Pydantic. The model never writes directly to SQL.

Human-in-the-Loop

Inventory changes require human approval. AI drafts invoices; humans confirm.

Defensive Guardrails

Food customizations are capped at 3 modifications.

Valid Transactions

If the ingredients of a dish are not fully available after the previous order, the dish will be listed as unavailable for subsequent customers.

Thank You!

Zero Canceled Orders
Eliminates out-of-stock cancellations.

Zero Wrong-Bag Claims
Physical QR validation stops driver mix-ups before leaving the store.

Merchant Health
Automates invoice bookkeeping and protects restaurant operating margins.